

Notes: Angrist & Lavy (QJE, 1999)

Using Maimonides' Rule to Estimate the Effect of Class Size on Scholastic Achievement

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1 Big picture

- **Question.** Does reducing class size raise student achievement? The paper estimates the causal effect of class size on Israeli elementary-school test scores.
- **Main problem.** Ordinary class-size comparisons are biased because class size is not randomly assigned. Smaller classes may be created for weaker students, disadvantaged schools, or remedial settings.
- **Key institutional rule.** Israeli public schools follow a maximum class size of 40, rooted in Maimonides' interpretation of the Talmud. If grade enrollment exceeds 40, the grade must be split into two classes; if it exceeds 80, into three; and so on.
- **Identification idea.** The rule creates a nonlinear, discontinuous relation between enrollment and class size. Around enrollments of 40, 80, 120, etc., class size drops sharply for reasons unrelated to individual students' unobserved ability.
- **Instrument.** The predicted class size generated by Maimonides' rule is used as an instrument for actual class size.
- **Design interpretation.** The strategy is a regression-discontinuity design: compare schools just below and just above enrollment cutoffs where the rule changes the number of classes.
- **Main result.** Smaller classes significantly improve test scores for fourth and fifth graders, especially in reading and for fifth-grade math. The evidence is much weaker for third graders.
- **Why OLS fails.** OLS estimates are small or even positive because schools may place weaker pupils into smaller classes or because class size is correlated with background variables.
- **Robustness.** The paper checks full samples, discontinuity subsamples, piecewise enrollment controls, dummy instruments based on cutoffs, pooled models, interactions with socioeconomic disadvantage, and third-grade data from a different year.
- **Economic takeaway.** Institutional assignment rules can create credible quasi-experimental variation. In this setting, the estimated effect of class size is educationally meaningful for fourth and fifth graders.

2 Institutional setting and data

2.1 Maimonides' rule

The rule says that a class should not exceed 40 pupils. If grade enrollment is e , the number of classes implied by the rule is

$$m(e) = \left\lfloor \frac{e-1}{40} \right\rfloor + 1. \quad (1)$$

The predicted class size is therefore

$$\text{fsc}(e) = \frac{e}{\lfloor (e-1)/40 \rfloor + 1}. \quad (2)$$

Interpretation: Predicted class size rises with enrollment until a cutoff is crossed, then falls abruptly when another class is added. For example, enrollment 40 implies one class of 40, while enrollment 41 implies two classes of about 20.5.

2.2 Why the rule is useful for identification

- Enrollment determines the number of classes through a mechanical formula.
- The rule creates sharp predicted class-size drops at enrollment cutoffs.

- If schools cannot precisely manipulate enrollment around these cutoffs, the resulting class-size variation is plausibly exogenous.
- The paper uses both the smooth variation in predicted class size and the discontinuities around thresholds.

Interpretation: The rule does not randomly assign individual students to classes. Instead, it creates quasi-random variation in average class size across grades and schools conditional on enrollment and controls.

2.3 Testing data

- The data come from a short-lived national testing program in Israeli elementary schools.
- In 1991, all fourth and fifth graders were tested in mathematics and Hebrew reading.
- In 1992, third graders were tested using a similar national program.
- Scores are scaled from 1 to 100 using composite scores from the exams.
- The main data are class-level averages rather than individual pupil scores.

Interpretation: The test-score outcome is the average performance in each class. The analysis treats the class as the unit of observation and corrects standard errors for within-school correlation.

2.4 School and class data

- Class size comes from administrative data collected during the school year.
- Enrollment is beginning-of-year grade enrollment.
- Controls include a school-level socioeconomic index called percent disadvantaged, or PD.
- Additional identifiers distinguish towns, Jewish public schools, religious/secular schools, and other administrative categories.
- The main sample excludes Arab schools and independent ultra-orthodox schools because of test-data and curriculum differences.

Interpretation: The main control PD is important because disadvantaged schools have different achievement levels and may also have different enrollments and resource allocations.

2.5 Samples

- The full 1991 samples contain about 2,000 classes in each of grades 4 and 5.
- The 1992 third-grade sample contains about 2,100 classes.
- The discontinuity samples restrict enrollment to windows around the rule's cutoffs, such as ± 5 or ± 3 pupils around 40, 80, 120, etc.

Interpretation: The full sample uses all rule-induced predicted class-size variation. The discontinuity sample focuses more narrowly on local comparisons around threshold points.

3 Measurement framework and estimands

3.1 Pupil-level model

The paper motivates the class-level analysis using a pupil-level model:

$$y_{isc} = X'_s \beta + n_{sc} \alpha + \mu_c + \eta_s + \varepsilon_{isc}, \quad (3)$$

where

- y_{isc} is the score of pupil i in class c and school s ;
- X_s is a vector of school characteristics, including PD and sometimes functions of enrollment;
- n_{sc} is class size;
- μ_c is a class-level component;
- η_s is a school-level component;
- ε_{isc} is pupil-level error.

Interpretation: The coefficient α is the causal effect of one more student in the class on achievement. A negative α means smaller classes improve scores.

3.2 Class-level equation

Averaging across pupils in the class gives the class-level equation:

$$\bar{y}_{sc} = X_s' \beta + n_{sc} \alpha + u_{sc}. \quad (4)$$

Interpretation: Because the data are class averages, the empirical regressions use average class scores as the dependent variable and actual class size as the endogenous regressor.

3.3 First stage

The first-stage relationship is

$$n_{sc} = \pi_0 + \pi_1 \text{fsc}(e_s) + X_s' \pi_2 + v_{sc}. \quad (5)$$

Interpretation: Maimonides' rule predicts actual class size strongly. The graphical evidence and Table III show that actual class size moves with fsc even after controlling for percent disadvantaged and enrollment.

3.4 Reduced form

The reduced form is

$$\bar{y}_{sc} = \gamma_0 + \gamma_1 \text{fsc}(e_s) + X_s' \gamma_2 + r_{sc}. \quad (6)$$

Interpretation: A negative reduced-form coefficient means that higher predicted class size is associated with lower achievement. This connects the institutional rule to student outcomes.

3.5 Instrumental variables estimand

The 2SLS coefficient is approximately

$$\hat{\alpha}_{IV} = \frac{\widehat{Cov}(\bar{y}_{sc}, \text{fsc}(e_s) \mid X_s)}{\widehat{Cov}(n_{sc}, \text{fsc}(e_s) \mid X_s)}. \quad (7)$$

Interpretation: The IV estimate scales the reduced-form effect of predicted class size on scores by the first-stage effect of predicted class size on actual class size.

4 Identification and empirical design

4.1 Regression-discontinuity logic

- The number of classes changes discontinuously when enrollment crosses 40, 80, 120, etc.
- Actual class size therefore falls sharply near each cutoff.
- If other determinants of achievement vary smoothly with enrollment, the discontinuities identify the causal effect of class size.

Interpretation: This is a fuzzy regression discontinuity because the rule predicts class size but does not perfectly determine actual class size.

4.2 Key identifying assumptions

1. **Continuity:** unobserved determinants of achievement vary smoothly with enrollment near the cutoffs.
2. **Relevance:** Maimonides' rule strongly predicts actual class size.
3. **Exclusion:** conditional on controls and smooth functions of enrollment, the rule affects test scores only through class size.
4. **No precise manipulation:** parents and schools do not sort precisely around enrollment thresholds in a way that changes unobserved student quality.

Interpretation: The most important threat is manipulation or sorting around cutoffs. The paper discusses parental avoidance and checks the credibility of the design using graphical patterns and discontinuity subsamples.

4.3 Why enrollment controls matter

- Enrollment itself may correlate with school size, community type, school resources, and student composition.
- The paper controls for enrollment linearly and with piecewise-linear functions.
- The discontinuity-sample design further reduces reliance on global functional form.

Interpretation: The rule-generated discontinuities are useful only if the smooth part of the enrollment-achievement relationship is controlled well enough.

4.4 Why OLS is not enough

- OLS class-size coefficients are small and often positive.
- This is consistent with endogenous class formation: weaker pupils may be placed in smaller classes.
- School principals may create small remedial classes or assign teachers nonrandomly.
- IV uses rule-induced class-size variation rather than observed class-size variation chosen by schools.

Interpretation: The contrast between OLS and IV is central to the paper. Ordinary comparisons do not capture the causal effect of reducing class size.

5 Results by section

5.1 I. Data and descriptive statistics

- The typical class has about 30 pupils.
- Average grade enrollment is about 78 pupils.
- Ten percent of classes have fewer than about 22 pupils and ten percent have more than about 37 pupils.
- Percent disadvantaged averages about 14 percent.
- Fourth- and fifth-grade score distributions look similar, while third-grade scores are higher and less variable, plausibly because the 1992 exams followed public controversy over the 1991 results.

Interpretation: The descriptive table shows substantial class-size variation and enough enrollment variation to exploit the rule.

5.2 II. Graphical analysis

- Figure I shows that actual class size closely follows the sawtooth pattern predicted by Maimonides' rule.
- Figures II and III show that test scores tend to fall when predicted class size rises.
- Residualized graphs control for percent disadvantaged and enrollment, making the discontinuity pattern more visible.

Interpretation: The figures provide visual evidence of both the first stage and the reduced form.

5.3 III. Measurement framework

- The model formalizes the class-size effect using a pupil-level achievement equation.
- Since data are class averages, the empirical model uses class-level average scores.
- The identification problem is that class size may be correlated with class-level and school-level unobservables.
- Maimonides' rule supplies an instrument for actual class size.

Interpretation: This section clarifies why the coefficient on actual class size cannot be interpreted causally without the rule-based instrument.

5.4 IV. Estimation results

- Table II shows OLS estimates. These are generally small and sometimes positive, suggesting selection bias.
- Table III shows the first-stage and reduced-form effects of predicted class size.
- Tables IV and V show 2SLS estimates for fifth and fourth graders in 1991.
- The strongest results are for fifth-grade reading/math and fourth-grade reading.
- Table VI shows that dummy instruments around discontinuity samples deliver qualitatively similar findings.
- Table VII pools grades and allows interactions with percent disadvantaged.
- Table VIII shows that third-grade estimates are weak and imprecise.

Interpretation: The IV evidence indicates that class-size reductions raise achievement in grades 4 and 5, but the effects are not uniform across all grades and subjects.

5.5 V. Characterizations and comparisons

- The authors compare their results with previous evidence, including randomized class-size experiments and observational studies.
- The magnitude is educationally meaningful: a reduction of several pupils can raise scores by a nontrivial fraction of a standard deviation.
- They also characterize the binary-instrument interpretation using discontinuity subsamples.

Interpretation: The paper positions Maimonides' rule as a quasi-experimental alternative to both randomized trials and ordinary observational regressions.

6 Tables and figures

6.1 Table I: Descriptive statistics

Read:

- Panel A summarizes the full samples for grades 5, 4, and 3.
- Average class sizes are about 30 pupils.
- Average enrollment per grade is about 78 pupils.
- Average percent disadvantaged is around 14 percent.
- Panel B focuses on discontinuity samples around enrollment cutoffs.

Economic message: The data contain rich class-size and enrollment variation, and the discontinuity samples provide local comparisons around the rule's thresholds.

TABLE I
UNWEIGHTED DESCRIPTIVE STATISTICS

Variable	Mean	S.D.	Quantiles				
			0.10	0.25	0.50	0.75	0.90
A. Full sample							
5th grade (2019 classes, 1002 schools, tested in 1991)							
Class size	29.9	6.5	21	26	31	35	38
Enrollment	77.7	38.8	31	50	72	100	128
Percent disadvantaged	14.1	13.5	2	4	10	20	35
Reading size	27.3	6.6	19	23	28	32	36
Math size	27.7	6.6	19	23	28	33	36
Average verbal	74.4	7.7	64.2	69.9	75.4	79.8	83.3
Average math	67.3	9.6	54.8	61.1	67.8	74.1	79.4
4th grade (2049 classes, 1013 schools, tested in 1991)							
Class size	30.3	6.3	22	26	31	35	38
Enrollment	78.3	37.7	30	51	74	101	127
Percent disadvantaged	13.8	13.4	2	4	9	19	35
Reading size	27.7	6.5	19	24	28	32	36
Math size	28.1	6.5	19	24	29	33	36
Average verbal	72.5	8.0	62.1	67.7	73.3	78.2	82.0
Average math	68.9	8.8	57.5	63.6	69.3	75.0	79.4
3rd grade (2111 classes, 1011 schools, tested in 1992)							
Class size	30.5	6.2	22	26	31	35	38
Enrollment	79.6	37.3	34	52	74	104	129
Percent disadvantaged	13.8	13.4	2	4	9	19	35
Reading size	24.5	5.4	17	21	25	29	31
Math size	24.7	5.4	18	21	25	29	31
Average verbal	86.3	6.1	78.4	83.0	87.2	90.7	93.1
Average math	84.1	6.8	75.0	80.2	84.7	89.0	91.9
B. +/- 5 Discontinuity sample (enrollment 36–45, 76–85, 116–124)							
	5th grade		4th grade		3rd grade		
	Mean	S.D.	Mean	S.D.	Mean	S.D.	
	(471 classes, 224 schools)		(415 classes, 195 schools)		(441 classes, 206 schools)		
Class size	30.8	7.4	31.1	7.2	30.6	7.4	
Enrollment	76.4	29.5	78.5	30.0	75.7	28.2	
Percent disadvantaged	13.6	13.2	12.9	12.3	14.5	14.6	
Reading size	28.1	7.3	28.3	7.7	24.6	6.2	
Math size	28.5	7.4	28.7	7.7	24.8	6.3	
Average verbal	74.5	8.2	72.5	7.8	86.2	6.3	
Average math	67.0	10.2	68.7	9.1	84.2	7.0	

Variable definitions are as follows: Class size = number of students in class in the spring, Enrollment =

6.2 Figures I–III: First stage and reduced form

Read:

- Figure I shows the first stage: actual class size follows the sawtooth pattern implied by Maimonides' rule.
- Figure II overlays test scores and predicted class size by enrollment.
- Figure III residualizes both scores and predicted class size on percent disadvantaged and enrollment, showing that the reduced-form pattern remains after controls.

Economic message: The graphical evidence shows the entire identification logic: the rule changes class size, and achievement moves in the opposite direction.

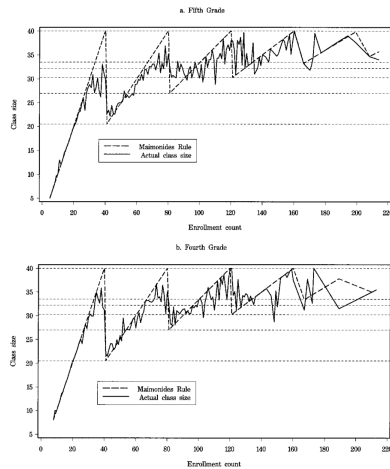


FIGURE I
Class Size in 1991 by Initial Enrollment Count, Actual Average Size and as Predicted by Maimonides' Rule

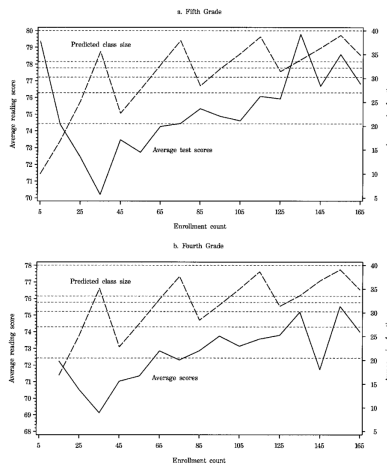


FIGURE II
Average Reading Scores by Enrollment Count, and the Corresponding Average Class Size Predicted by Maimonides' Rule

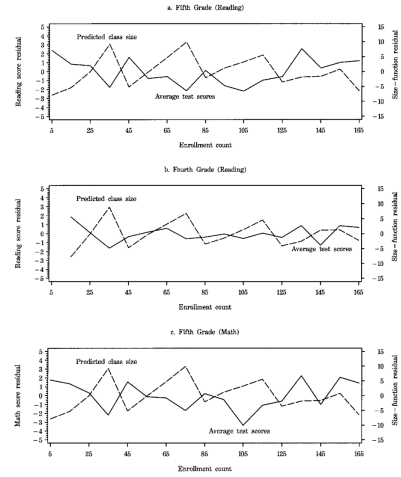


FIGURE III
Average Test (Reading/Math) Scores and Predicted Class Size by Enrollment, Residuals from Regressions on Percent Disadvantaged and Enrollment

(a) Figure I: actual and predicted class size.

(b) Figure II: scores and predicted class size.

(c) Figure III: residualized scores and predicted class size.

6.3 Table II: OLS estimates for 1991

Read:

- OLS estimates for fifth and fourth graders are small and generally not negative in a stable way.
- Adding percent disadvantaged substantially increases explanatory power.
- Enrollment controls do not convert the OLS estimates into a clear causal pattern.

Economic message: OLS comparisons do not provide credible causal evidence because observed class size is chosen by schools and correlated with student composition.

	5th Grade						4th Grade					
	Reading comprehension			Math			Reading comprehension			Math		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Mean score (s.d.)	74.3 (8.1)			67.3 (9.9)			72.5 (8.0)			69.9 (8.8)		
Regressors												
Class size	.221 (.031)	-.031 (.026)	-.025 (.031)	.322 (.036)	.076 (.044)	.019 (.033)	-.053 (.028)	-.040 (.033)	.221 (.036)	.055 (.033)	.009 (.039)	
Percent disadvantaged	-.350 (.012)	-.351 (.013)	-.340 (.018)	-.332 (.018)	-.332 (.018)	-.332 (.018)	-.339 (.013)	-.341 (.014)	-.289 (.016)	-.281 (.016)	-.282 (.016)	-.282 (.016)
Enrollment			-.002 (.006)		.017 (.009)			-.004 (.007)		.014 (.008)		
Root MSE	7.54	6.10	9.36	8.32	8.30	7.94	6.65	8.66	7.82	7.81	7.81	7.81
R ²	.036	.369	.309	.048	.249	.252	.013	.309	.025	.204	.207	.207
N	2,019			2,018			2,049			2,049		

The unit of observation is the average score in the class. Standard errors are reported in parentheses. Standard errors were corrected for within-school correlation between classes.

	5th Graders						4th Graders					
	Class size		Reading comprehension		Math		Class size		Reading comprehension		Math	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
A. Full sample												
Means	29.9 (6.3)		74.4 (7.7)		67.3 (9.6)		30.3 (6.3)		72.5 (8.0)		68.9 (8.8)	
Regressors												
f _{cs}	.704 (.022)	.542 (.027)	-.111 (.028)	-.149 (.035)	-.009 (.039)	-.124 (.049)	.772 (.020)	.670 (.025)	-.085 (.031)	-.089 (.040)	.038 (.037)	-.033 (.047)
Percent disadvantaged	-.076 (.010)	-.053 (.009)	-.360 (.012)	-.355 (.013)	-.354 (.017)	-.338 (.018)	-.054 (.008)	-.039 (.009)	-.340 (.013)	-.340 (.014)	-.292 (.016)	-.282 (.016)
Enrollment			.045 (.005)	.010 (.006)	.021 (.009)	.027 (.009)			.001 (.007)	.001 (.007)	.019 (.009)	
Root MSE	4.56	4.38	6.07	6.07	8.33	8.28	4.20	4.13	6.64	6.64	7.83	7.81
R ²	.516	.553	.375	.377	.247	.255	.561	.575	.311	.311	.204	.207
N	2,019		2,019		2,018		2,049		2,049		2,049	
B. Discontinuity sample												
Means	30.8 (7.4)		74.5 (8.2)		67.0 (9.2)		31.1 (7.2)		72.5 (7.8)		68.7 (8.7)	
Regressors												
f _{cs}	.481 (.051)	.346 (.052)	-.197 (.050)	-.202 (.054)	-.089 (.071)	-.154 (.077)	.625 (.050)	.503 (.053)	-.075 (.056)	.059 (.063)	.012 (.072)	
Percent disadvantaged	-.130 (.029)	-.067 (.028)	-.424 (.029)	-.422 (.029)	-.435 (.039)	-.405 (.042)	-.068 (.029)	-.029 (.028)	-.348 (.032)	-.343 (.034)	-.306 (.041)	-.291 (.043)
Enrollment			.006 (.015)	.003 (.015)	.041 (.022)	.041 (.022)			.007 (.017)	.007 (.017)	.034 (.022)	
Root MSE	5.95	5.58	6.24	6.24	8.58	8.53	5.49	5.26	6.57	6.57	8.26	8.25
R ²	.360	.437	.421	.421	.296	.296	.428	.428	.299	.299	.178	.182
N	471		471		471		415		415		415	

The function f_{cs} is equal to $\text{enrollment}/(\text{enrollment} - 1,040 + 1)$. Standard errors are reported in parentheses. Standard errors were corrected for within-school correlation between classes. The unit of observation is the average score in the class.

6.4 Table III: Reduced-form estimates for 1991

Read:

- Predicted class size fsc is strongly positively related to actual class size.
- For fifth graders, fsc is negatively related to reading and math scores.
- For fourth graders, the strongest reduced form is for reading.

- Results are shown for both the full sample and the discontinuity sample.

Economic message: Table III establishes both instrument relevance and the reduced-form relation between rule-induced class size and achievement.

6.5 Tables IV–V: 2SLS estimates for fifth and fourth graders

Read:

- Table IV reports fifth-grade 2SLS estimates using Maimonides’ predicted class size as an instrument.
- Fifth-grade estimates are negative and often statistically significant, especially for reading and math specifications with controls.
- Table V reports fourth-grade estimates.
- Fourth-grade reading effects are negative; fourth-grade math effects are weaker and less stable.
- Full-sample and discontinuity-sample estimates tell a similar qualitative story.

Economic message: The IV estimates suggest that smaller classes improve achievement for older elementary students, especially fifth graders and fourth-grade reading.

	Reading comprehension				Math			
	Full sample		+/- 5 Discontinuity sample		Full sample		+/- 5 Discontinuity sample	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mean score (s.d.)		74.4 (7.7)				74.5 (8.2)		67.3 (9.6)
Regressors								
Class size	-.158 (.040)	-.275 (.066)	-.260 (.081)	-.186 (.104)	-.410 (.113)	-.582 (.181)	-.013 (.056)	-.230 (.092)
Percent disadvantaged	-.372 (.014)	-.369 (.013)	-.369 (.013)	-.477 (.037)	-.461 (.037)	-.355 (.019)	-.350 (.019)	-.261 (.049)
Enrollment	.022 (.009)	.012 (.026)			.053 (.028)	.041 (.012)	.062 (.037)	-.459 (.049)
Enrollment squared/100			.005 (.011)			-.010 (.016)		-.435 (.036)
Piecewise linear trend				.136 (.032)				.193 (.040)
Root MSE	6.15	6.23	6.22	7.71	6.79	7.15	8.34	8.40
N		2019		1961	471		2018	1960

The unit of observation is the average score in the class. Standard errors are reported in parentheses. Standard errors were corrected for within-school correlation between classes. All estimates use f_{cs} as an instrument for class size.

(a) Table IV: fifth graders.

	Reading comprehension				Math			
	Full sample		+/- 5 Discontinuity sample		Full sample		+/- 5 Discontinuity sample	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Mean score (s.d.)		72.5 (8.0)				72.5 (7.8)		67.3 (9.6)
Regressors								
Class size	-.110 (.040)	-.133 (.059)	-.074 (.067)	-.147 (.084)	-.098 (.090)	-.150 (.128)	.049 (.048)	-.050 (.070)
Percent disadvantaged	-.346 (.014)	-.345 (.014)	-.346 (.014)	-.354 (.034)	-.347 (.017)	-.290 (.017)	-.284 (.017)	-.098 (.042)
Enrollment	.005 (.008)	-.040 (.024)			.017 (.022)	-.020 (.010)	.007 (.029)	-.299 (.043)
Enrollment squared/100			.021 (.011)				.006 (.014)	.095 (.028)
Piecewise linear trend				.100 (.020)				.130 (.028)
Root MSE	6.65	6.66	6.63	8.02	6.64	6.69	7.82	7.82
N		2049		2001	415		2049	2001

The unit of observation is the average score in the class. Standard errors are reported in parentheses. Standard errors were corrected for within-school correlation between classes. All 2SLS estimates use f_{cs} as an instrument for class size.

(b) Table V: fourth graders.

6.6 Table VI: Dummy-instrument discontinuity estimates

Read:

- The paper replaces the continuous predicted-class-size instrument with binary instruments based on being above or below cutoffs.
- The estimates use discontinuity samples around enrollment thresholds.
- The pattern remains broadly consistent with the main IV results, though precision varies across subject and grade.

Economic message: The dummy-instrument approach emphasizes local discontinuities and reduces reliance on the full functional form of Maimonides’ rule.

6.7 Table VII: Pooled estimates and disadvantaged interactions

Read:

- The table pools fourth and fifth graders and estimates class-size effects across grades.

TABLE VI
DUMMY-INSTRUMENT RESULTS FOR DISCONTINUITY SAMPLES

	5th grade						4th grade					
	Reading comprehension			Math			Reading comprehension			Math		
	+/- 5	+/- 3	+/- 5	+/- 3	+/- 5	+/- 3	+/- 5	+/- 3	+/- 5	+/- 3	+/- 5	+/- 3
	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample	Sample
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
<i>Regressors</i>												
Class size	-.687 (.197)	-.588 (.198)	-.451 (.236)	-.596 (.254)	-.395 (.254)	-.270 (.281)	-.175 (.130)	-.234 (.157)	-.380 (.205)	.018 (.162)	-.118 (.202)	-.247 (.234)
Percent disadvantaged	-.464 (.039)	-.452 (.045)	-.433 (.050)	-.416 (.058)	-.350 (.034)	-.372 (.043)	-.291 (.043)	-.323 (.055)				
Segment 1 (enrollment 36-45)	-5.09 (2.40)	-4.54 (2.59)	-10.7 (3.19)	-7.54 (3.07)	-6.94 (3.34)	-12.6 (3.80)	-1.62 (1.77)	-2.67 (2.23)	-6.94 (2.90)	-1.89 (2.21)	-3.57 (2.87)	-7.31 (3.31)
Segment 2 (enrollment 76-85)	-1.64 (1.41)	-2.18 (1.64)	-2.96 (2.00)	-1.57 (1.83)	-2.17 (2.14)	-2.89 (2.41)	-1.52 (1.24)	-2.16 (1.59)	-3.83 (2.10)	-1.15 (1.56)	-2.50 (2.07)	-3.96 (2.39)
Root MSE	7.46	7.24	8.67	9.41	9.14	10.2	6.72	6.70	8.30	8.25	8.53	9.52
N	471	302	471	302	415	265	415	265	415	265	415	265

The table reports results from a sample of classes in schools with enrollment close to points of discontinuity. The unit of observation is the average score in the class. Standard errors are reported in parentheses. Standard errors were corrected for within-school correlation between classes. All estimates use $1/f_c < .32$ and interactions with dummies for enrollment segments as instruments for class size. Since there are three segments, there are three instruments. The models include dummies for the first two segments to control for segment main effects.

TABLE VII
POOLED ESTIMATES AND MODELS WITH PERCENT DISADVANTAGED INTERACTION TERMS

	5th grade		4th grade		Pooled estimates			
					Reading		Math	
	Reading	Math	Reading	Math				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Regressors</i>								
Class size	-.156 (.074)	-.080 (.104)	-.101 (.067)	.019 (.080)	-.197 (.047)	-.120 (.054)	-.127 (.061)	-.019 (.070)
Percent disadvantaged	-.162 (.068)	-.091 (.094)	-.288 (.073)	-.162 (.086)	-.356 (.012)	-.222 (.056)	-.315 (.015)	-.126 (.071)
Grade 4					-1.93 (.158)	-1.89 (.160)	1.52 (.193)	1.57 (.194)
Enrollment	.018 (.009)	.036 (.012)	.004 (.008)	.018 (.010)	.013 (.007)	.010 (.007)	.029 (.009)	.026 (.009)
<i>Interaction</i>								
Class size*PD	-.008 (.003)	-.010 (.004)	-.002 (.003)	-.005 (.003)		-.005 (.002)		-.007 (.003)
Root MSE	6.25	8.43	6.66	7.82	6.44	6.44	8.10	8.11
N	2019	2018	2049	2049	4068	4068	4068	4067

The unit of observation is the average score in the class. Standard errors are reported in parentheses. Standard errors were corrected for within-school correlation between classes. All estimates use f_c and f_c^* PD as instruments for class size and class size*PD.

- It also allows the effect to vary with percent disadvantaged.
- The pooled estimates retain negative class-size effects for reading and, in some specifications, math.

Economic message: The pooled models show that the basic result is not purely a single-grade artifact and explore whether class-size effects differ by socioeconomic composition.

6.8 Table VIII and Figure IV: Third graders and binary-instrument interpretation

Read:

- Table VIII reports results for third graders tested in 1992.
- The third-grade estimates are imprecise and do not provide clear evidence of beneficial class-size effects.
- Figure IV shows CDFs of class size in the discontinuity sample by the value of a binary instrument.
- The CDFs illustrate how the instrument shifts the distribution of class size.

Economic message: The third-grade results show the limits of the evidence. The strongest causal findings are for fourth and fifth graders in 1991.

TABLE VIII ESTIMATES FOR THIRD GRADERS									
	Reading comprehension					Math			
Class size	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	RF	OLS	RF	IV	IV	OLS	RF	IV	IV
Mean score (s.d.)					86.3 (6.1)				84.1 (6.8)
Regressors									
Class size		-.020 (.027)		-.052 (.047)	-.040 (.055)	.023 (.032)		-.005 (.056)	-.068 (.065)
Percent disadvantaged	-.044 (.009)	-.176 (.011)	-.175 (.011)	-.177 (.012)	-.177 (.012)	-.110 (.013)	-.112 (.013)	-.112 (.014)	-.110 (.013)
Enrollment	.019 (.005)	.0004 (.005)	.002 (.006)	.003 (.006)	-.006 (.021)	.006 (.006)	.008 (.007)	.008 (.008)	.058 (.025)
Enrollment squared/100					.004 (.007)				-.023 (.008)
f_w	.691 (.025)		-.036 (.033)				-.003 (.038)		
Root MSE	4.19	5.67	5.67	5.67	5.67	6.63	6.63	6.63	6.63
R^2	.546	.144	.144			.056	.056		

The unit of observation is the average score in the class. Standard errors are reported in parentheses. Standard errors were corrected for within-school correlation between classes. There are 2111 third grade classes. The RF column heading denotes reduced-form estimates.

(a) Table VIII: third graders.

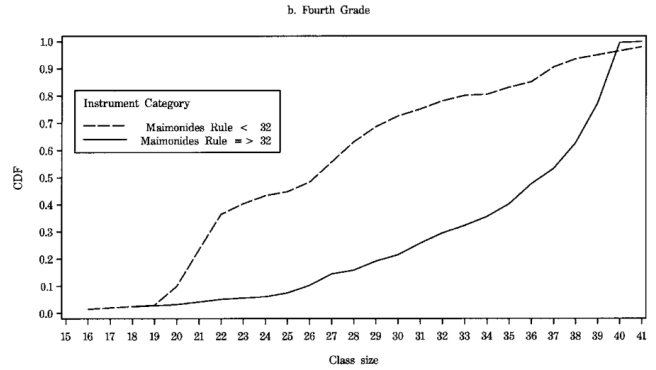
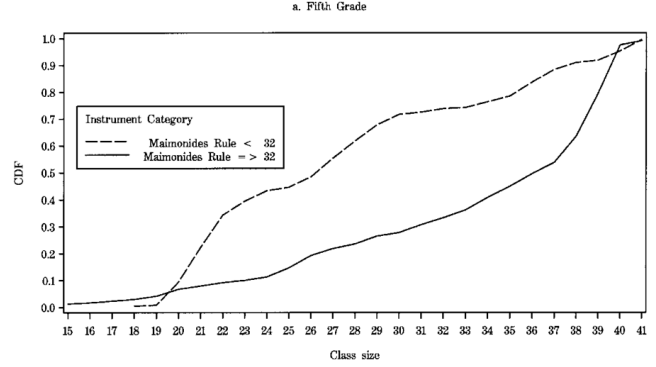


FIGURE IV
CDFs of Class Size in the ± 5 Discontinuity Subsample, Separately by Value of a Binary Instrument Based on Maimonides' Rule

(b) Figure IV: CDFs by binary instrument.

7 Short summary

- The paper estimates the causal effect of class size using Maimonides' maximum class-size rule of 40.
- The rule implies predicted class size $fsc(e) = e/(\lfloor(e-1)/40\rfloor + 1)$.
- Predicted class size creates discontinuities at enrollments 40, 80, 120, etc.
- The data come from Israeli national tests for fourth and fifth graders in 1991 and third graders in 1992.
- OLS class-size estimates are small or positive, consistent with nonrandom class formation.
- Maimonides' rule provides a strong first stage for actual class size.
- Reduced-form estimates show negative associations between rule-predicted class size and scores, especially for fifth graders and fourth-grade reading.
- 2SLS estimates imply that smaller classes improve fourth- and fifth-grade achievement.
- Third-grade estimates are weak and imprecise.
- The paper is a classic application of regression-discontinuity ideas to education production.

8 Conclusion

8.1 Core conclusion

Angrist and Lavy show that Maimonides' rule generates quasi-experimental variation in class size. Using this rule as an instrument, they find that smaller classes raise test scores for fourth and fifth graders, while ordinary OLS comparisons fail to reveal this effect.

8.2 Economic intuition

The intuition is that class size is usually endogenous. Schools may create smaller classes for weaker pupils or disadvantaged settings, biasing simple comparisons. Maimonides' rule changes class size mechanically through enrollment thresholds. Around those thresholds, class-size variation is more plausibly unrelated to unobserved student ability.

8.3 Takeaway

The paper's lasting contribution is methodological and substantive. Methodologically, it demonstrates how an institutional rule can be used as a regression-discontinuity instrument. Substantively, it provides credible evidence that class-size reductions can improve achievement, at least for fourth and fifth graders in this setting.